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Air-borne diseases

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The objectives of this lecture are to

- 1- Identify the program of controlling of air-borne diseases
- 2- Explain the occurrence of diphtheria disease ,its signs and symptoms, diagnosis ,treatment, control and prevention.
- 3- Explain epidemiology of mumps disease , its occurrence and distribution, clinical features , treatment, control and prevention

- Many bacteria and viruses invade the body through the **respiratory tract**.
- These microorganisms can affect the **upper and lower respiratory tract** and some of them can spread to other organs as well.
- Upper respiratory infections are **very common**.
- Acute lower respiratory infections are **a major cause of mortality in children** in the developing countries.

→ Thankfully less common

Major air-borne infections include the following:

Viral infections

- Mumps *→ endemic in our country*
- Measles *→ spring disease*
- Rubella (German measles)
- Influenza
- Infectious mononucleosis
- Chickenpox

Bacterial

- TB *↪ endemic in our country*
- Pneumonia (pneumococcus, streptococcus, haemophilus, klebsiella, mycoplasma)
- Whooping cough *↪ Pertussis* (Bordetella pertussis)
- Streptococcal infections (tonsillitis, rheumatic fever)
- Meningitis (Neisseria meningitidis)
- Diphtheria (Clostridium diphtheriae) *↪ winter disease*
- Plague, pneumonic

Fungal

- Histoplasmosis (histoplasma capsulatum)

→ Opportunistic infection with chronic immunosuppressive patient

Transmission of air-borne infections

Air-borne microorganisms are transmitted to the new host through three methods:

1. ^{→ direct} **Droplets**: particles of saliva and mucous secretions are ejected by coughing, sneezing, talking, laughing and spitting.
- These droplets may contain food debris and microorganisms.



- Since the droplets are **heavy** they tend to settle to the ground so transmission with this route only occurs at **very short distance**. *→ half a meter*
- Droplets are also **larger** than being capable of entering the **lower respiratory tract**. *↳ if distance is large droplet will fall to ground and becomes useless. / so only upper respiratory tract*



2. Droplet nuclei:

- When droplets evaporate **before** settling on the ground
- Dry nucleus remains which is small and light
- Hence easily carried away by air and capable of **entering the bronchioles and alveoli.**

↳ enter lower respiratory tract

↑ travel long distance



3. **Dust**: some organisms can persist for a long time in dust which then acts as **a reservoir**. *non-living*
- These organisms may be derived from **sputum or settled droplets**.

** kinda all three methods come from droplet*



Control of air-borne diseases

Main lines of controlling these disease can be grouped under three headings:

- ❖ The agent
- ❖ Mode of transmission
- ❖ The host.

1. Elimination of the agent:

The causative organism can be reduced and eliminated through

- .Elimination of the human or animal reservoir
- .Elimination of dust, cleanliness and disinfection of floors and interior of dwellings



2. Interruption of transmission: transmission of the agent to a new host can be minimized through the following measures:

- Air hygiene by
 - Good **ventilation** of living places
 - **Disinfection** of the air with ultraviolet light in contaminated places.

→ air flowing in and out reducing stagnant air (also)

- Avoidance of **overcrowding** at home, in working places and outdoors.
- **Personal hygiene:**
 - Avoidance of **coughing, talking and sneezing** directly to the face of other people.
 - Use of **masks** ^{of tissue or elbow (common sense)} by patients can limit spread of the microorganism to others.



3. Protection of the host: individual can be protected from air-borne infections by

- **Specific immunization** in case of diseases for which a specific vaccine is available such as measles, influenza and whooping cough or by using immunoglobulins such as for measles.

(active immunity)

live-attenuated

live (Attenuated), subunit

Peptides or whole subunit

Passive immunity



- Chemoprophylaxis:

Using medication to prevent disease in high risk people such as **isoniazid** for prevention of **TB**.

also set line in TB higher dose than chemoprophylaxis (800 mg)

Diphtheria



- An **acute** bacterial infection
- involving the **tonsils, pharynx, nose and larynx**
- Caused by bacillus **corynebacterium diphtheria** which liberates a **cytotoxin**. *the disease is caused by this toxin that bacteria release not the bacteria itself*
- **Asymptomatic** cases(colonization of the bacteria on the mucosa) are much **more** common than **symptomatic** cases.



The characteristic feature of the disease is

① fever → but heart rate increase exponentially in relation to fever

presence of **whitish membrane** adherent to the mucosa with surrounding inflammation.

② it will be difficult to remove, diff color, whit white color, cause bleeding when removed since it's necrotic tissue
whitish in streptococcal it's easy to remove and (bright white)

Sore throat and cervical lymphadenopathy are common.

h
in sever cases cause bull's neck and require immediate tracheostomy

Nasal diphtheria may cause nasal discharge and obstruction.

In **moderate and severe** cases there is swelling of the neck with extensive membrane formation in the trachea which may lead to **airway obstruction**.



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“Bull neck” appearance of diphtheritic cervical lymphadenopathy



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The ^{cytotoxin} toxin may cause myocarditis, heart block and congestive heart failure.

Neuropathy and skeletal muscle weakness can happen later.

Case fatality rate is 5-10%.



Diagnosis:

Clinical diagnosis is based on the presence of the **membrane** in a patient with upper respiratory infection and lymphadenopathy.

Confirmation is by **microscopy** and **culture** of the lesion.

→ characteristic & fine exponential febrile between HR-TEMP

to dx
↳ after foreign swab
immediates GUL embolic anti-toxin ± antibiotic



Occurrence

Diphtheria is the disease of **colder** months that happens among children or adults who have **not** been **vaccinated**.

A massive **outbreak** started in Russia in 1990 and spread to all countries 1990-1997.

32

more than 150,000 cases and 5000 deaths were reporting during this period.

Reservoir: human *only*

Mode of transmission: **air-borne** through contact with a **patient** or a **carrier** or rarely contact with items contaminated with the lesions.

Contaminated milk can serve as a vehicle for the bacteria. *not reservoir basically sick people cough on it and bacteria is simply just surviving in it*

Incubation period: 2-5 days

Latent period very short people becomes infectious on day of infection

Period of infectiousness:

As long as virulent bacteria are discharged from the lesions, usually **2 weeks**.

Antibiotics can quickly stop shedding of the bacteria.

Very rarely carriers can shed the bacteria for as long as **6 months**.

Control

a. Preventive measures:

A. Education of the public about danger of diphtheria and need for vaccination of children.

Vaccination: *→ in PENTA and DPT* vaccination is the best and most effective way for preventing diphtheria.

A toxoid vaccine is available and part of routine immunization of children.



Diphtheria vaccine is given along with tetanus toxoid and killed pertusis vaccine (DTP or triple vaccine) to all children.

PENTA

In Iraq three doses are given 2-months apart starting from age of 2 months.

DPT

Booster doses are advice during the second *(18 month)* year and before school. *(4-6 years)*

b. Control of patients and contacts:

Notification: Reporting all cases to health authorities, Isolation of the patients until 2 of each throat and nose cultures are negative.

*also consecutive
diagnosis*

If culture facility is not available isolation should continue for 2 weeks of antibiotic therapy.

+ anti-toxic

Disinfection of all items used or contaminated by the patient.

Isolation of contacts: contacts who work in **food handling** or who **mix with un-immunized** children must **stay away** from their work until they are managed as below.

Management of contacts: all close contacts of patients must have **cultures** of throat and nose taken and kept under surveillance for **one week**.



-All household contacts must receive a single dose of **benzathine penicillin IM 600,000 IU less than 30kg. 2.4 million IU over 30kg.** *or* **Azthromycine** in children 10-12 mg/kg once daly In adult 500mg once daly or one week oral **erythromycin** regardless of their immunization status.

-Food handlers and those working with non-immunized children must stay away from work.



- in addition to the other vaccines*
- Previously immunized contacts must get a **booster dose** of diphtheria toxoid if **5 years** have lapsed since the last dose. *like pneumococcal vaccine*
 - Non-immunized contacts must get **a new series of the vaccine.** *usually 3*



Specific treatment: ^{in diseased} diphtheria **antitoxin** should be given as soon as diagnosis is suspected and bacteriological specimens are taken **without waiting for results.** ^(essential)

Antibiotics such as **penicillin or azithromycin**
benzathine penicillin IM 600,000 IU less than 30kg.
2.4 million IU over 30kg ^{single dose}

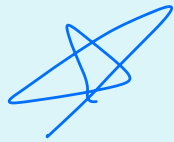


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Azithromycine in children 10-12 mg/kg once daly
In adult 500mg once daly 10-14 day

must also be given but this is not a substitute
for the antitoxin.

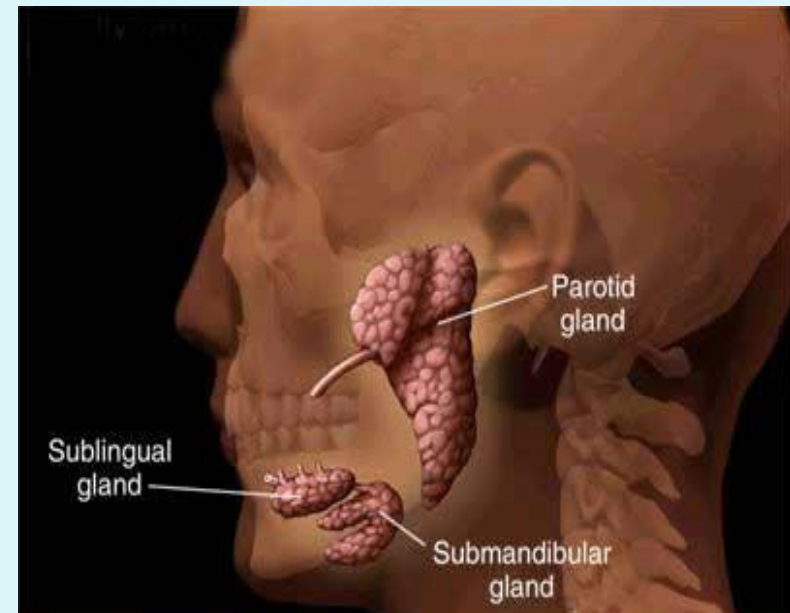
mandal tar



MUMPS



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Epidemiology

1-Mumps was common until the mumps vaccine was licensed in 1967.

2-Before the vaccine more than 200,000 cases occurred each year in the United States.

Since then the number of cases has dropped to fewer than 1,000 a year,

3-**Epidemics** have become **fairly rare**

4-As in the prevaccine , most cases of mumps are still in children **ages 5 to 14**,

5- Mumps infections are uncommon in children younger than **1 year old**.

Caused by:

Mumps virus is a member of the **Paramyxoviridae** family of viruses.

Incubation period: 14–24 days.

Period of communicability: 6 days before
Symptoms to 6 after the symptoms started

Swelling of parotid gland

Transmission:

1-direct contact or by droplets spread by the
airborne route.

2-indirect , any contact of saliva



Occurrence and distribution

Mumps is probably more common than assumed to be with up to **85%** of the population found to have been infected in adult life

although few would have manifested the disease.

→ before introduction of vaccine

→ way too much asymptomatic cases

With such a high proportion of the population meeting the virus,

there is a relatively high risk of complications and cost–benefit studies have shown that vaccination produces substantial economic savings.



Clinical features

It is an infection of the salivary glands producing enlargement and pain in the parotid gland, *also could be affected by bacteria* *Bacteria: hard salivary gland on palpation* but can lead to orchitis, *Mumps: Spongy swelling when palpated -> it will firm up when you put finger through it* mastitis, *inflammation of mammary gland* meningitis, pancreatitis and acute respiratory symptoms .



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Commonly an infection of children 2–5 years of age.

The more serious manifestations are more likely in adults, especially males.

→ occur especially when there's vaccine failure

Cases of mumps may start with

- 1-A fever of up to 39.4° Celsius
- 2-Headache
- 3-Loss of appetite.
- 4-The well-known hallmark of mumps is swelling and pain in the parotid glands
- 5-The glands usually become increasingly swollen and painful over a period of 1 to 3 days.



6-The pain gets worse when the child swallows, talks, chews, or drinks acidic juices

Start unilateral then progress to be bilateral
7-Both the left and right parotid glands may be affected, with one side swelling a few days before the other only one side may swell.

8- In rare cases, mumps will attack other groups of salivary glands instead of the parotids.

9-Mumps in adolescent and adult males may also result in the development of orchitis, (inflammation of the testicles).

Usually one testicle becomes swollen and painful about 7 to 10 days after the parotids swell.

This is accompanied by
high fever
shaking chills
headache
nausea & vomiting
and abdominal pain that can sometimes be
mistaken for appendicitis if the right testicle
is affected.

*pain in RLQ → orchitis no rebound tenderness
→ Appendicitis rebound tenderness +ve*



After 3 to 7 days,
testicular pain and swelling subside,
usually at about the same time that the
fever passes.

In some cases, both testicles are involved.
Even with involvement of both testicles,
sterility is only a rare complication of
orchitis.

→ still one testes enlarge sores



10- Mumps may affect the pancreas in females, the ovaries, causing pain and tenderness in parts of the abdomen. *→ in both genders*
Right ovary may be confused with appendicitis

11-In some cases, signs and symptoms are so mild that no one suspects mumps infection.

12-1 in 3 people may have a mumps infection without symptoms.

13-Because mumps can also involve the brain and its membranes,
if it cause meningitis stiff neck, convulsions (seizures), extreme drowsiness, severe headache, or changes of consciousness.



Diagnosis:

is made on **clinical grounds**, but
serological confirmation can be made with
mumps-specific IgM, a rise in IgG
or culture of saliva or urine.

Hx + examination

Treatments

1-Because mumps is caused by a virus, it cannot be treated with antibiotics.

2-Fever subsided by **paracetamole** ,
acetaminophen or ibuprofen
These drugs will also help relieve pain.

Handwritten note: Patient may not
over acid hypersensitivity
Just home isolation

3- Soothing the child's swollen parotid glands
with **either warm or cold packs**.

4-Serve a soft, bland diet that does not require a lot of chewing and encourage the child to drink plenty of fluids.

5- Avoid serving tart or acidic fruit juices that make parotid pain worse.

Water, decaffeinated soft drinks, and tea are better tolerated.



6-In orchitis medications for pain and swelling , instructions on how to apply warm or cool packs to soothe the area and how to provide extra support for the testicles.

7-A child may play quietly. *→ since talking worsen pain*



Continuous vomiting
high, high fever
complications (e.g. orchitis, meningitis, mastitis)

8- Hospitalization in sever cases

a-Supportive treatment

- IV fluid
- Antipyretic (Paracetamole)

b-In secondary infection give antibiotics

Control and prevention

1-Notification to LHO *→ Central Health Authority*

2-Health Education

3- Isolation of the patients

4-Treatment of the patients

5-Immunization

Mumps ,measles and rubella(MMR) vaccine.
However, if less than 75% of children are vaccinated,
then this could result in an epidemiological shift to older age groups,
increasing the likelihood of complications.



If mumps vaccination is included,
then a second opportunity,
either by the routine program
or by catch-up campaigns, should be given
unless coverage is over 90%.

if 90% and above we have herd immunity and at this stage there's no need for catch-up campaign

Two doses of the MMR vaccine are recommended before a child enters school: *by WHO differ than iraq*

- The first in 15 months
- The second between ages 4 and 6 years,
- or between 11 and 12 if not previously given

In response to a mumps outbreak in the Midwest, college students and health care workers in particular are encouraged two doses of the MMR vaccine.

A single dose doesn't appear to offer sufficient protection during an outbreak.

Side effects of the vaccine

^{usually (normally) red}
1-Can't get mumps from the MMR vaccine,

2- mostly no side effects from the vaccine.

3-A few cases with a mild fever or rash, and some people (mostly adults) have joints pain afterward for a short time.

4- Less than one out of a million doses causes a serious allergic reaction.

→ Live attenuated → hence can't be used
→ immunocompromised
→ pregnancy
→ > 50 years
→ chronic steroid
→ HIV



Contraindication of the vaccine :

- 1-**Pregnant** women or women who plan to get pregnant within the next four weeks
- 2-People who have had a life-threatening **allergic reaction to the antibiotic neomycin** *so be careful if gravimetallic reaction*
- 3-People with **compromised immune systems** .

Although concerns have been raised about a connection between the **MMR vaccine** and **autism**,
extensive reports from the

- ❖ American Academy of Pediatrics,
- ❖ Institute of Medicine
- ❖ Centers for Disease Control and Prevention



conclude that there's **no scientifically proven link** between the MMR vaccine and autism.

In addition, there's no scientific benefit in separating these vaccines.

→ making separate individual components
→ measles
→ mumps
→ rubella

These organizations note that autism (autism spectrum disorder) is often identified in toddlers between the ages of 18 and 30 months, which happens to be about the time children are given their first MMR vaccine.

But this coincidence in timing shouldn't be mistaken for a cause-and-effect relationship.

→ NO SCIENTIFIC EVIDENCE
JUST COINCIDENCE